



ECUE 422 - Réglementation Certification et interopérabilité des SIH

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“Standard” HL7

Autorité publique indépendante à caractère scientifique dotée de la personnalité morale, responsable pour :

Évaluation et recommandation : d'un point de vue médical et économique les produits, actes, prestations et technologies de santé, en vue de leur remboursement.

Accréditation et certification : certifie les établissements de santé, elle accrédite les praticiens de certaines disciplines médicales sur la base du volontariat. Elle participe également à l'amélioration de la qualité de l'information médicale sur internet et dans la presse, elle certifie la visite médicale, ainsi que les logiciels d'aide à la prescription.

La HAS conçoit et met également à disposition des acteurs de santé des outils, guides et méthodes afin d'améliorer leur prise en charge ou la mise en œuvre de leurs projets.

Grille de pilotage et de sécurité d'un projet de télémédecine (juin 2013) [[lien](#)]

Efficiencia de la télémédecine - état des lieux de la littérature internationale et cadre d'évaluation [[lien](#)]

La donnée de sante

Lorsqu'on parle des données de santé, il y a en réalité plusieurs types:

Administrative

Financière

Clinique

Le problème : **absence de normalisation !**

Plusieurs solutions possibles, peu retenues.

Health Level 7 (HL7)

HL7 (<http://www.hl7.org/>) : - Association des EUA à but non-lucratif

Ensemble de standards permettant l'échange d'informations entre plusieurs systèmes de santé

Organisation à la charge de ces standards (1987).

HL7 versions 2 / 3 / FHIR et OpenEHR

Définit comment les données seront enregistrées et échangées

HL7 Version 2

Un exemple :

Cet exemple est issu de Google HealthCare API
(<https://cloud.google.com/healthcare/docs/concepts/hl7v2>)

```
1 MSH|^~\&|FROM_APP|FROM_FACILITY|TO_APP|TO_FACILITY|20180101000000||ADT^A01
2 |20180101000000|P|2.5|
3 EVN|A01|20110613083617|
4 PID|1|843125^^^MRN|21004053^^^MRN~2269030303^^^ORGNMBR||SULLY^BRIAN
5 ||19611209|M|||123 MAIN ST^CITY^STATE^12345|
6 PV1||I|H73 RM1^1^^HIGHWAY 01 CLINIC|||5148^MARY QUINN|
7 |||||||Y|||||||||||||||||||||||20180101000000|
```

HL7 Version 3

Basé sur XML :

```
<?xml version="1.0" encoding="UTF-8"?>
<PRPA_IN101001UV01 ITSVersion="XML_1.0" xmlns="urn:hl7-org:v3"
  xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance">
  <id extension="3948375" root="2.16.840.1.113883.19.10.700363.2288"/>
  <creationTime value="20060501140010"/>
  <versionCode code="NE2006"/>
  <!-- Interaction is a notification of a person registration -->
  <interactionId extension="PRPA_IN101001UV01" root="2.16.840.1.113883.1.6"/>
  <processingCode code="P"/>
  <processingModeCode code="T"/>
  <acceptAckCode code="ER"/>
  <receiver>
    <device>
      <id extension="922" root="2.16.840.1.113883.19.9"/>
      <name>Master MPI</name>
      <asAgent>
        <representedOrganization>
          <id extension="1002003" root="2.16.840.1.113883.19.200"/>
          <name>Alpha Hospital</name>
        </representedOrganization>
      </asAgent>
    </device>
  </receiver>
  <sender>
    <device>
      <id extension="1" root="2.16.840.1.113883.19.9"/>
    </device>
  </sender>
```


Fast Healthcare Interoperability Resources (FHIR)

<http://hl7.org/fhir/>

Standard utilisé pour l'échange de données médicales
publié par l'organisation internationale HL7

Utilisateurs majeurs : NHS (Roy. Uni) ; Apple, Amazon, Google,
Microsoft

Open source

Le formatage des fichiers se fait en JSON, XML, ND-
JSON, RDF

Possibilité d'échanges :

RESTful API (HTTP) REST requests.

Messaging – messages entre systèmes

Documents – échanges de documents

Exemple en JSON

```
"active": true,  
"name": [  
  {  
    "use": "official",  
    "family": "Chalmers",  
    "given": [  
      "Peter",  
      "James"  
    ]  
  },  
  {  
    "use": "usual",  
    "given": [  
      "Jim"  
    ]  
  },  
  {  
    "use": "maiden",  
    "family": "Windsor",  
    "given": [  
      "Peter",  
      "James"  
    ],  
    "period": {  
      "end": "2002"  
    }  
  }  
],
```

Open industry specifications, models and software for e-health

<https://www.openehr.org/>

Open-source concurrent du FHIR

Dans le cadre de ce cours, nous n'aborderons pas ce format



HL7 Version 2

Health Level 7:

“Standard” for health data communications

➤ <http://www.hl7.org>

➤ Interoperability

➤ optimize workflow

➤ reduce ambiguity

Health Level 7:

Does not provide:

- Security/Access Control – HL7 does not provide for the enforcement of a user's security policies. In addition, HL7 does not specify a specific encryption method.
- Privacy/Confidentiality – HL7 does not address this issue and makes no assumption about how the data will be used at the source or destination of a message.
- Accountability/Audit trails – HL7 does not attempt to define possible transaction processing features needed in a user's environment.

MSH|^~\&|EPIC|EPICADT|SMS|SMSADT|199912271408|CHARRIS|ADT^A
04|1817457|D|2.5|
PID||0493575^^^2^ID
1|454721||DOE^JOHN^^^^|DOE^JOHN^^^^|19480203|M||B|254 MYS-
TREET AVE^^MYTOWN^OH^44123^USA|| (216)123-
4567||M|NON|400003403~1129086|
NK1||ROE^MARIE^^^^|SPO|| (216)123-
4567||EC||||||||||||||||||||||
PV1||O|168~219~C~PMA^^^^^^^^|||277^ALLEN MYLASTNAME^BON-
NIE^^^^||||||||
||2688684|199912271408|||||002376853

Health Level 7:

Definitions:

- HL7 – an application protocol for electronic data exchange in healthcare environments
- **Message** – the atomic unit of data transferred between systems, consisting of segments in a defined sequence
 - **Message Type** – the specific purpose
- **Segment** – a logical grouping of data fields identified by three letter identification (MSH, PID, OBX, ...)
- **Field** – a string of characters defined by one of the HL7 data types, inside a segment
 - **Data Type** – a restriction on the contents of a field
 - **Field Separator** – character to separate two adjacent data fields within an

Health Level 7:

Definitions:

- **Repetition** – a field with multiple values
 - **Repetition Separator** – character to separate two adjacent data fields within an HL7 segment
- **Component** – splits a field in sub-parts

Health Level 7:

Definitions:

- **Order** – a request for material or service
- **Observation** – performance of the service including result data
- **Composite** : field
- **Trigger Event** – a real world event that initiates an exchange of messages. There is a one to many relationship between message type and trigger event.

Health Level 7:

Rules:

- Message Header Segment (MSH) is required and always first
- There may be more than one type of segment and can be nested
 - [] – Optional segment
 - { } – Repeating segment

Optionality	
R – Required	RE – Required but may be empty
O – Optional	C – Conditional depending on trigger event or some other field
X – not used with this trigger event	B – Backwards compatible with previous HL7 versions

Health Level 7:

Rules:

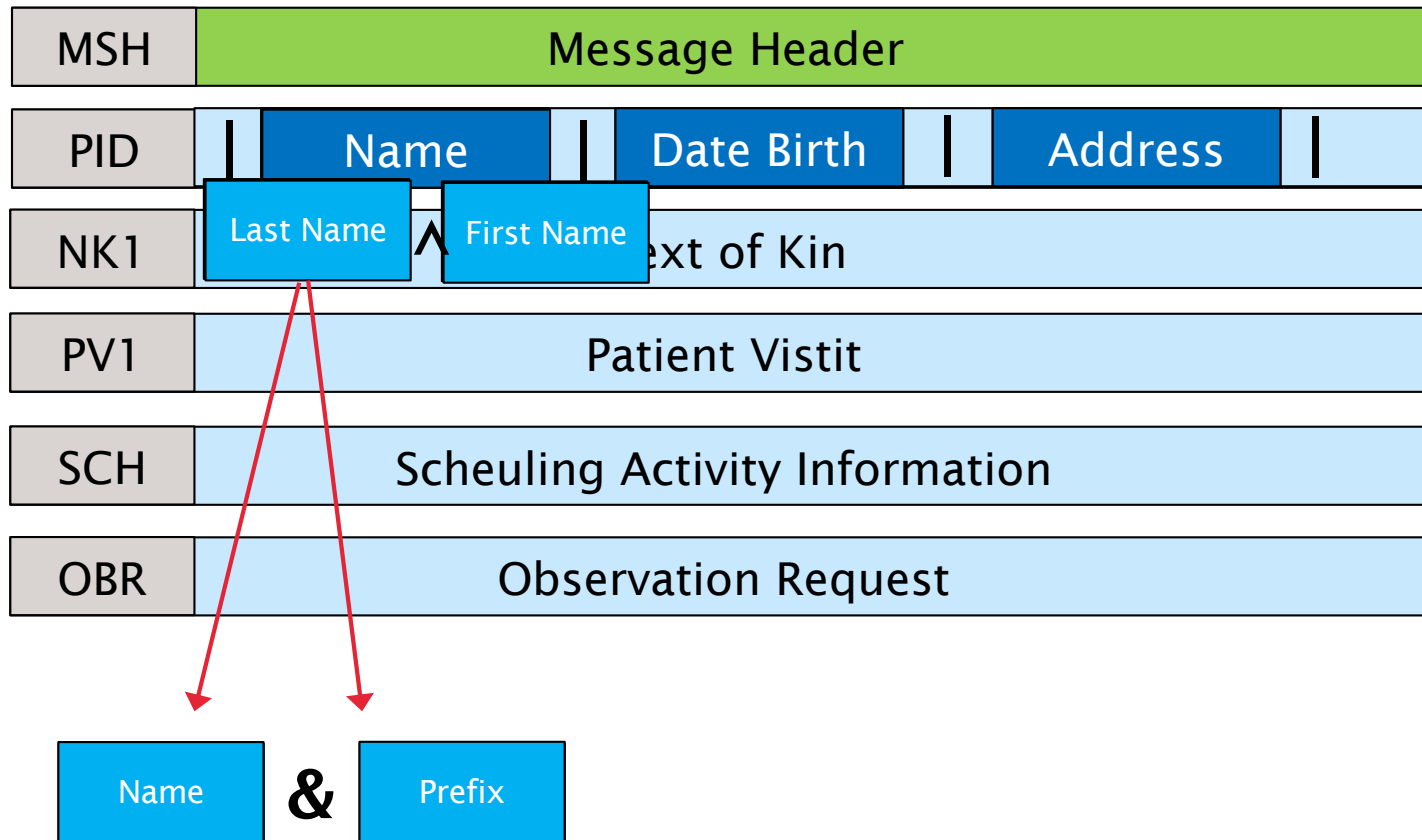
- Recommended message delimiters (field 2 of MSH)

Delimiter	Value
Field Separator	
Component Separator	^
Subcomponent Separator	&
Repetition Separator	~
Escape Character	\

HL7 Message

MSH	Message Header
PID	Patient Identification
NK1	Next of Kin
PV1	Patient Visit HL7 Message
SCH	Scheduling Activity Information
OBR	Observation Request
Header	Segment

HL7 Message



HL7 Message

PID		Name		Date Birth		Address	
-----	--	------	--	------------	--	---------	--

```
|MSH|^~\&|ECUE431|Instance1||201904141725||ADT^A04|1817457|D|2.5.1|123456||AL  
|PID||0797675^^^^MR|454721||Stark^Robb^^^^B|Tully^Catelyn^^^^|19900401|M|  
King's Road^^Winterfell^North^^Westeros||S|AGN|546446849~66784|  
|NK1||Stark^Catelyn^^^^|MTH|| (33)666545447||NC|||||||
```

Comment savoir où insérer les informations ? Documentation !

HL7 Message

PID		Name		Date Birth		Address	
-----	--	------	--	------------	--	---------	--

```
|MSH|^~\&|ECUE431|Instance1||201904141725||ADT^A04|1817457|D|2.5.1|123456||AL  
|PID||0797675^^^^MR|454721||Stark^Robb^^^^B|Tully^Catelyn^^^^|19900401|M|  
King's Road^^Winterfell^North^^Westeros||S|AGN|546446849~66784|  
|NK1||Stark^Catelyn^^^^|MTH|| (33)666545447||NC|||||||
```

PID: Patient Identification

The PID segment is used by all applications as the primary means of communicating patient identification information. This segment contains permanent patient identifying and demographic information that, for the most part, is not likely to change frequently.

HL7 Message

```
|MSH|^~\&|ECUE431|Instance1||201904141725||ADT^A04|1817457|D|2.5.1|123456||AL
|PID||0797675^MR|454721|Stark^Robb^B|Tully^Catelyn^|19900401|M|
King's Road^Winterfell^North^Westeros||S|AGN|546446849~66784|
|NK1|Stark^Catelyn^|MTH|| (33)666545447||NC|||||||
```

SEQ	LENGTH	DT	OPT	RPT / #	TBL #	NAME
PID.1	4	<u>SI</u>	O	1		Set ID - PID
PID.2	20	<u>CX</u>	B	1		Patient ID
PID.3	250	<u>CX</u>	R	*		Patient Identifier List
PID.4	20	<u>CX</u>	B	*		Alternate Patient ID - PID
PID.5	250	<u>XPN</u>	R	*		Patient Name
PID.6	250	<u>XPN</u>	O	*		Mother's Maiden Name
PID.7	26	<u>TS</u>	O	1		Date/Time of Birth
PID.8	1	<u>IS</u>	O	1	0001	Administrative Sex
PID.9	250	<u>XPN</u>	B	*		Patient Alias
PID.10	250	<u>CE</u>	O	*	0005	Race
PID.11	250	<u>XAD</u>	O	*		Patient Address
PID.12	4	<u>IS</u>	B	1	0289	County Code
PID.13	250	<u>XTN</u>	O	*		Phone Number - Home
PID.14	250	<u>XTN</u>	O	*		Phone Number - Business
PID.15	250	<u>CE</u>	O	1	0296	Primary Language
PID.16	250	<u>CE</u>	O	1	0002	Marital Status
PID.17	250	<u>CE</u>	O	1	0006	Religion
PID.18	250	<u>CX</u>	O	1		Patient Account Number

HL7 Message

```
|MSH|^~\&|ECUE431|Instance1||201904141725||ADT^A04|1817457|D|2.5.1|123456||AL
|PID||0797675^^^^MR|454721|Stark^Robb^^^^B|Tully^Catelyn^^^^|19900401|M|
King's Road^^Winterfell^North^^Westeros|||S|AGN|546446849~66784|
|NK1||Stark^Catelyn^^^^|MTH|||(33)666545447||NC|||||||
```

SEQ	LENGTH	DT	OPT	TBL #	NAME
XPN.1	194	<u>FN</u>	O		Family Name
XPN.2	30	<u>ST</u>	O	<u>FirstName</u>	Given Name
XPN.3	30	<u>ST</u>	O		Second And Further Given Names Or Initials Thereof
XPN.4	20	<u>ST</u>	O		Suffix (e.g., Jr Or Iii)
XPN.5	20	<u>ST</u>	O		Prefix (e.g., Dr)
XPN.6	6	<u>IS</u>	B	<u>0360</u>	Degree (e.g., Md)
XPN.7	1	<u>ID</u>	O	<u>0200</u>	Name Type Code
XPN.8	1	<u>ID</u>	O	<u>0465</u>	Name Representation Code
XPN.9	483	<u>CE</u>	O	<u>0448</u>	Name Context
XPN.10	53	<u>DR</u>	B		Name Validity Range
XPN.11	1	<u>ID</u>	O	<u>0444</u>	Name Assembly Order
XPN.12	26	<u>TS</u>	O		Effective Date
XPN.13	26	<u>TS</u>	O		Expiration Date
XPN.14	199	<u>ST</u>	O		Professional Suffix

Health Level 7:

Acknowledgement messages are sent to indicate if receiving application was able to:

- Parse message
- Decode message
- Assume responsibility for the message
- Process message contents
- Successfully commit to storage

Health Level 7:

Rules : Null versus empty

- Null is used when data is present but without a value “ “
- The field is left blank when there is no data

- Example: PID | | |54321|9876-3|Doe^John|”“|...

Rules : Receiving systems:

- Will ignore the data within segments, fields, etc that are present but not expected
- Will treat expected segments that are not present as a segment with all fields not present
- Will treat expected fields and components that are not included as not present

HL7 – Common message types

Type	Abbreviation
Admission, Discharge, Transfer	ADT
General Clinical Order Message	OMG
Imaging Order	OMI
Laboratory Order Message	OML
Pharmacy/treatment Order Message	OMP
General Order Message	ORM
Unsolicited Observation/Result	ORU
Unsolicited Vaccination Record Update	VXU

HL7 – Common data types

Type	Abbreviation	Definition
Coded Element	CE	Used to transmit codes and associated text
Composite Quantity with units	CQ	Specifies the numeric quantity or amount, and the units in which the quantity is expressed
Extended Address	XAD	Specifies the address of a person, place, or organization plus associated information
Extended Person Name	XPN	Specifies the complete name of a person plus associated information
Numeric	NM	A number (integer)
String Data	ST	Any displayable/printable ACSII characters intended for strings less than 200 characters
Text Data	TX	Longer string data intended for display purposes
Time Stamp	TS	Specifies a point in time including time zone

HL7 – Common data types

```
MSH|^~\&|LABGL1||DMCRES||199812300100||ORU^R01|LABGL1199510221838581|P|2.3
|||NE|NE
PID|||6910828^Y^C8||Newman^Alfred^E||19720812|M|W|25 Centscheap Ave^^
Whatmeworry^UT^85201^^P|||(555) 777-6666|(444) 677-7777||M||773789090
OBR||110801^LABGL|387209373^DMCRES|18768-
2^CE)^LN|||199812292128||35^ML|||||IN2973^Schadow^Gunther^^^^MD^UPIN
||||||^Once|||||CA20837^Spinosa^John^^^^MD^UPIN

OBX||NM|4544-3^HEMATOCRIT (AUTOMATED)^LN||45||39-49
|||F|||199812292128||CA20837
OBX||NM|789-8^ERYTHROCYTES COUNT (AUTOMATED)^LN||4.94|10*12/mm3
|4.30-5.90|||F|||199812292128||CA20837
LL COUNTS+DIFFERENTIAL TESTS (COMPOSITE
```

Segments

MSH: Message Header

PID: Patient Identification

OBR: Observation Request

OBX: Observation Result

Delimiters

- | Field
- ^ Component
- & Subcomponent
- ~ Repetition
- \ Escape Character

Documents

- <https://hl7-definition.caristix.com/v2/>
– (All documentation. Very good!)
- <https://www.hl7soup.com/hl7tutorials.html>
- <https://www.youtube.com/watch?v=ZAQdYR1rmEQ>
-



Python HL7

<https://python-hl7.readthedocs.io/en/latest/>

- Message is a **list**
 - Every element is a segment
- Every segment can be accessed as a nested list
- Similar to Fields, Components, ...
- Messages can also be **dictionaries!**

HL7

```
10 # The HL7 structure, as represented in this code:
11 #
12 # Message[segment][field][repetition][component][sub-component]
13 #
14 # https://python-hl7.readthedocs.io/en/latest/
15
16 import hl7
17
18 #ATTENTION : we don't start a line with delimiter '|'
19 message = """MSH|^~\&|ECUE431|Instance1||201904141725||ADT^A04|1817457|D|
20 2.5.1|123456||AL\r"""
21 message += """|PID||0797675^^^^MR|454721||Stark^Robb^^^^B|Tully^Catelyn^^^^|
22 #19900401|M|King's Road^^Winterfell^North^^Westeros||S|AGN|546446849~66784|\r"""
23 message += """|NK1|Stark^Catelyn^^^^MTH||(33)666545447|NC|||||||||\r"""
24
25 h = hl7.parse(message) # creates a hl7.Message object,
26                        # wrapping a series of hl7.Segment objects
27 print(type(h)) # print type
28 print(str(h)) # converts hl7 to string and print it
29
30 print(len(h)) #the length is the number of segments
```

```
31
32 print('\n\n1>>>')
33 print(h[1])
34 print('2>>>')
35 print(h[1][0])
36
37 print('3>>>')
38 print(h[1][5])
39 print('4>>>')
40 print(h[1][5][0])
41 print('5>>>')
42 print(h[1][5][0][1])
43
44 print('6>>>')
45 print(h.segments('PID'))
46 print('7>>>')
47 print(h.segments('PID')[0])
48 print('8>>>')
49 print(h.segments('PID')[0][5][0][1])
50
51
52 #check objects
53 print("\n\nTypes of objects")
54 print(type(h)) #Message
55 print(type(h[1])) #Segment
56 print(type(h[1][5])) #Field
57 print(type(h[1][5][0])) #Repetition
58 print(type(h[1][5][0][1])) #Component
```